IoT based Smart Vehicle Presence Sensor SPIN-V for Smart Parking System

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***Abstract*—Currently, smart cities are the result of the need to direct our lives towards sustainability through the development of solid strategic planning that uses infrastructure, innovation, and technology to adapt to changes brought about by the overpopulation of cities. Due to the growing vehicle fleet in the cities, the management of parking spaces has become very important and with it the interest in developing smart parking systems that promote the efficient and intensive use of information and communication technologies. This work presents the design and development of a smart vehicular presence sensor (SPIN-V, in its Spanish acronym), which is the cornerstone of a smart parking system (SEI-UVM, in its Spanish acronym). The SPIN-V is composed by a distance sensor and a camera to detect the entry of a vehicle to the parking space, and a LED indicator to inform that space has been reserved through a mobile application. Information on the occupation of the parking spaces is sent in real time to the monitoring center. The SPIN-V allows the efficient management of parking spaces generating a saving of time in the search while reducing vehicular traffic.**

***Index Terms*—Smart parking system, smart city, internet of thing.**

I. INTRODUCTION

Nowadays, the implementation of Internet of things (IoT) techniques has grown due to its several applications in all no scientific areas such as medicine, agriculture, social sciences, computer sciences, and no scientific areas, like government, society, industry, among others. IoT is expressed into diverse areas that are classified according to the problems that can be solved, for instance: health, agriculture, smart cities, networks, among others. Smart cities is a concept that allows to integrating technologies into a community with the aim to make life easier with a minimal effort.

Smart cities are the result of the need to guide our lives toward sustainability. The initiatives of smart and sustainable cities do not have to be seen as a model of a distant and unattainable future, but as a necessity to the current reality to face the challenges of the current societies where information and communication technologies (ICT) would play a transversal role as articulating tools that guarantee, and where appropriate facilitate, social cohesion, security, and sustainability [2]. These cities use infrastructure, innovation, and technology to adapt the changes brought about by the overpopulation of cities and thus contribute to the reduction of carbon dioxide emissions, decrease in energy consumption, and promoting economic, social and environmental development. It is worthwhile to mention that one of the first steps that gets implemented to upgrade the traditional cities to become smart cities is the smart parking [1].

Many cities have already started smart parking projects. Intelligent parking helps drivers to search, efficiently and effectively, for parking spaces through information and communication technology [4]. Intelligent parking spaces are systems that optimize the way of parking, either by streamlining the process or by reducing the space required. In this way, we seek to have a more respectful attitude towards the environment [3].

In [7] an architecture for smart parking is proposed by means of a mobile application connected to the cloud. The system provides parking facilities in real time and users are able to reserve places and make payments before to arrive at the parking space. There exists other systems that use wireless communication to reserve places and recommendations of nearest parking spaces through the global positioning system (GPS), [10]. The system transmits the availability of spaces every two minutes, if there is some parking space occupied no actions are made, in another case, any user is able to reserve a place within 2 km of their location. The GPS coordinates are received by any smartphone and the user receive a message with directions. The application is not functional if there is no WIFI connection and no one action is performed if another person is stationed in the parking space.

An intelligent parking algorithm is presented in [6], which calculates the optimal parking place for the user based on the distance of trajectory and time. It should be noted that this system does not have the reservation service and is subject to the availability of the place at that time. On the other hand, the smart parking system proposed in [9], presents the management of parking spaces in real time using the cloud as a means of communication and database. The system has a mobile application to make space reservations and an ultrasonic sensor placed on the ground connected via Ethernet.

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A novel parking system designed for smart cities is proposed in [13]. The system is connected to the cloud and the guidance function is based on the ant colony algorithm to calculate the shortest path between two points, but the authors do not present any application. There exist some proposals which focus their efforts in the available technologies for smart parking spaces. In [11], sensors, technologies, and interfaces used to collect and display real time parking occupancy information is proposed. Some applications are compared and the results show that the parking occupancy sometimes cannot be displayed due to the lack and the number of users connected to the applications. The authors explain that all the existing smart parking technologies and applications are not suitable for open parking lots due to varying environmental conditions and high expenditure.

Definitely, these current smart parking systems allow you to park a vehicle quickly and efficiently. However, they require a special infrastructure that is expensive in terms of installation, time, and money. The foregoing raises an area of innovation opportunity in the automation of private parking lots looking for practicality in its installation, modularity, and design, which may trigger the introduction of new business models. Therefore, this work presents the design and development of the intelligent vehicular presence sensor (SPIN-V, in its Spanish acronym), which is the cornerstone of the intelligent parking system (SEI-UVM, in its Spanish acronym).

II. National Observatory of Smart Environments

Research centers, universities, and industry have gathered efforts to collaborate on technological development projects. The universities have taken place in projects with industry and government. One of them is the largest private universities in Mexico, the Universidad del Valle de México (UVM) which considers the research as one of the keys of education; with over than 30 campuses around Mexico, 179,073 high school, master and university students, 8,726 teachers, more than 60 researchers, 200,000 graduate students and 5,974 administration staff (UVM, 2018) is responsible for the creation of a Research Center of Investigation, Innovation and Technological development (CIIDETEC-UVM Centro de Investigación, Innovación y Desarrollo Tecnológico - UVM in Spanish) since 2009.

CIIDETEC-UVM took the first step in the creation of a smart space called *"National Observatory of Smart Environments (OBNiSE)"*. The objective of OBNiSE is integrating strategies, methods, methodologies, prototypes, and in turn, monitoring the behavior of new systems and devices connected to an internal network to provide an integrated system and services. The observatory is equipped with electronic and electrical devices that allow the monitoring of the SEI-UVM system, explained as follows.

III. Smart Parking System SEI-UVM

The proposed SEI-UVM is a smart parking system that is devoted to private parking lots and it is composed of three main elements: the SPIN-V, a mobile application, and a monitoring center (see Fig. 1). Each parking place is equipped with a SPIN-V located in the middle of the back end of the parking place, the mobile application is for the driver/user to monitor and reserve a parking space developed and monitored in the OBNiSE, and finally the monitoring center is for the owner of the private parking lots to manage and control the parking spaces.

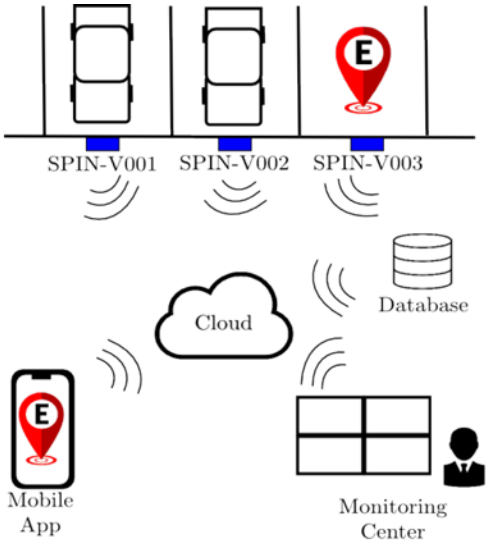


Fig. 1. Smart parking system SEI-UVM.

A. Network architecture of the Smart Parking System SEI-UVM

The SEI-UVM can be used as a management and control system of private parking lots, in which the availability of the parking places are shown in real time to the driver/user through a mobile app. In the mobile app, the user can enter a profile and register his/her vehicle's plates, monitor the available parking places in real time, reserve a parking space and obtain different route recommendations to get to the parking lot. On the other hand, the SPIN-V involves a distance sensor to detect the entry of a vehicle to the parking place, a camera to take an image of the vehicle's plate, a LED indicator and a buzzer to inform the status of the parking space. Moreover, the information on the occupation of the parking spaces is displayed in real time to the monitoring center.

The proposed network architecture of the smart parking system SEI-UVM is depicted in Fig. 2. Moreover, a more detailed explanation about the performance of the complete smart parking system is carried out below.

When a new vehicle is sensed by the proximity distance sensor of the SPIN-V, it takes an image of the vehicle's plate and performs an image processing algorithm to convert from image to text so the plate's number can be sent to the Assignation Manager. On the other hand, the mobile app displays the parking place status: free/reserved/occupied/unavailable, so the user can reserve or release a previous parking place reservation. If a new parking space is booked by the user, then this notification is sent to the Assignation Manager, so it

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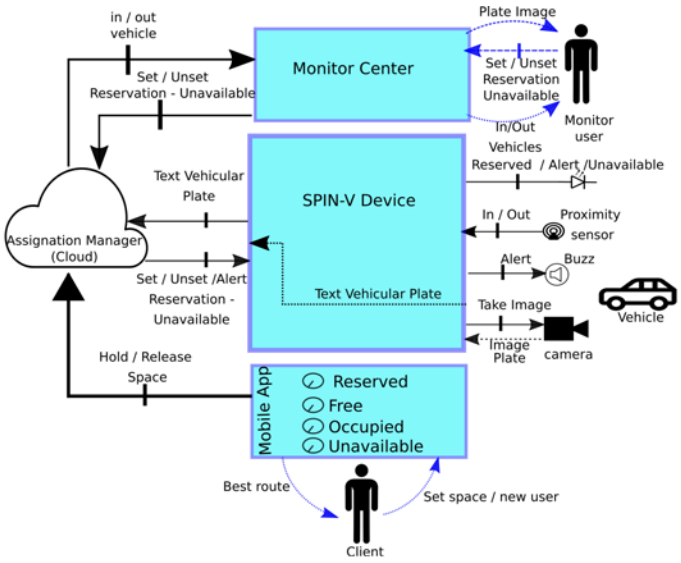


Fig. 2. Network architecture of the SEI-UVM.

can set the SPIN-V LED to the correspondent status: Red → Reserved, Off → Free and modify the parking place status on the monitor center.

A sequence diagram of the SEI-UVM is depicted in Fig. 3 in order to show the object interactions arranged in time sequence, where the objects and classes involved in the scenarios and the sequence of messages exchanged between the objects needed to carry out the functionality of three scenarios. This diagram can be associated with the use case realizations in the logical view of the proposed system.

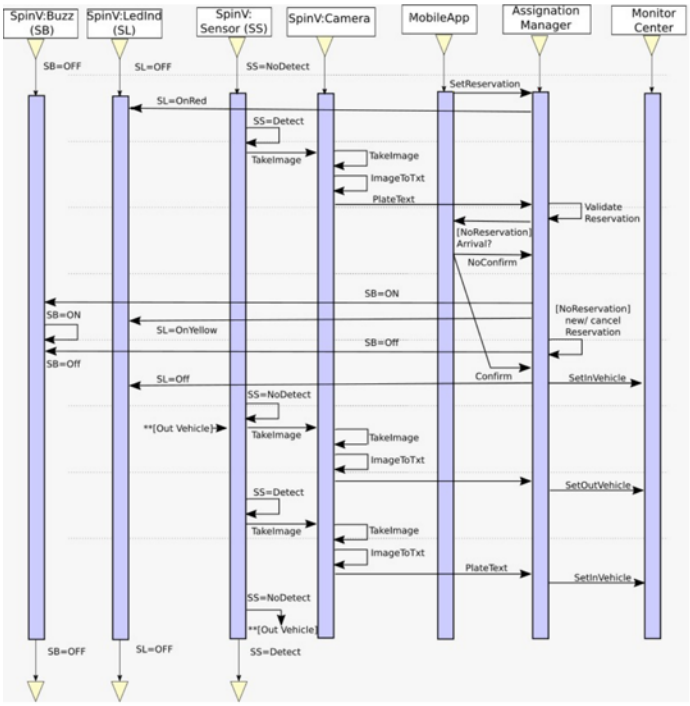


Fig. 3. Sequence diagram of the SEI-UVM.

- **Scenario 1:** The driver/user set a reservation via the mobileApp and send the requirement to the Assignation Manager, and set on red the SpinV:LedInd of the reserved parking place, so when the SpinV:Sensor detects a new vehicle takes the vehicle plate image (SpinV:Camera), and internally get the text of the image vehicle plate, which is sent to the Assignation Manager and stored. The cloud informs the monitor service of the new parked vehicle and validates if the image plate is equal to the reservation, asking to the reservation owner via mobileApp if it is correct, if not is confirmed (NoConfirm) then the SpinV:LedInd is set to yellow and the SpinV:Buzz sounds, until a new reservation or cancellation is executed.
- **Scenario 2:** The driver/user set a reservation via the mobileApp and send the requirement to the Assignation Manager, and set on red the SpinV:LedInd of the reserved parking place, so when the SpinV:Sensor detects a new vehicle take the vehicle plate image (SpinV:Camera), and internally get the text of the image vehicle plate, which is sent to the Assignation Manager and stored. The cloud informs the monitor service of this new parked vehicle and validates if the image plate is equal to the reservation, if yes (confirm) the SpinV:LedInd is set off.
- **Scenario 3:** The SpinV:LedInd is off and a driver/user arrives to the parking place, so when the SpinV:Sensor detects his/her vehicle take the vehicle plate image (SpinV:Camera), and internally get the text of the image vehicle plate, which is sent to the Assignation Manager and stored. The cloud informs the monitor service of this new parked vehicle.

Despite the scenario, when the SpinV:Sensor detects a vehicle is leaving the parking space the SpinV:Camera take the vehicle plate image, and then the plate vehicle text is sent to the cloud, which informs the monitor of the out of the vehicle.

IV. Design and Implementation of the IoT Sensor SPIN-V

This work presents the design and development of the IoT based sensor SPIN-V, which is the mainstay of the smart parking system (SEI-UVM) and represents the first multidisciplinary project carried out in the National Observatory of Smart Environments.

According to the logical view of the smart parking system SEI-UVM, the design considerations for the SPIN-V IoT sensor should contemplate the modularity, the reduction in time and economic investment in the installation and maintenance; which will give to the IoT sensor the flexibility to offer several business models that could be applied for commercial and residential uses (see Fig. 4).

A. Design of the IoT Sensor SPIN-V

The SPIN-V is designed to be implemented with the following components: the Raspberry pi 3 electronic card, an HC-SR04 ultrasonic sensor, the Pi v2 camera, a LED strip, a buzzer

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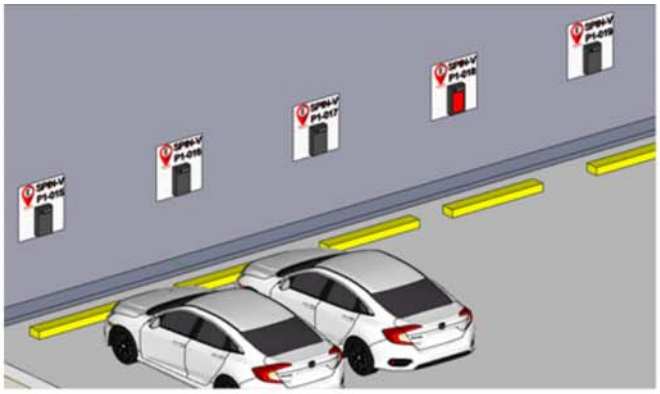


Fig. 4. SPIN-V installed in the back end of the parking space either for a commercial or residential use.

and a battery. The description of each of the components is shown below.

- The Raspberry Pi is a small computer which consists of a motherboard on which a processor, graphics chip, RAM is mounted. It is responsible for controlling the sensors, the led indicator, the Pi Camera, and to managing the communication between the overall system and the cloud services, so the system can read and write data in them.
- An ultrasonic sensor HC-SR04 allows us to measure distances thanks to its two transducers, a piezoelectric emitter and a receiver. The emitter emits pulses of ultrasound at a frequency of 40KHz, if an object is found, the sound waves will bounce. The amount of time it takes to return the wave determines the distance. The sensor is used to sense the presence of a vehicle in a parking space.
- A Pi V2 camera which offers high-speed, high-sensitivity images, is used to identify automobile plates.
- A LED strip is used in the physical system of the SPIN-V to indicate to the driver the status of the parking place. Within the secondary elements of the SPIN-V is the battery that is the power of the Raspberry Pi.
- A buzzer is an electroacoustic transducer that produces a continuous sound or buzzing of the same tone (usually acute) and it serves as a signaling or warning mechanism.
- A battery an electrochemical cells that transform chemical energy into electricity and it powers all components.

The connection diagram of the components of the SPIN-V is shown in (Fig. 5).

1) Design of the vision and proximity system for the IoT Sensor SPIN-V: Vision systems are based on digital sensors within industrial cameras with optics specialized in acquiring images, so that hardware and software can process, analyze and measure different characteristics to make decisions.

The Pi V2 camera is capable of performing automatic control functions with high performance OmniBSI technology (high sensitivity, low crosstalk, low noise). And as mentioned previously, it is used to take the license plates of the car, when the presence of a car was detected in the parking space.

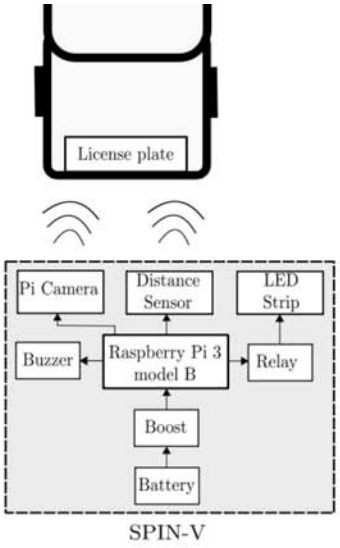


Fig. 5. Connection diagram SPIN-V.

To define the best position, orientation of the Pi camera, it is necessary to take into account the following concepts in order to get the best out of the field of view of the camera. The fundamental parameters for the imaging system are as follows [5]:

- Field of Vision (FOV): The visible area of the object.
- Working Distance (WD): The distance from the front of the lenses to the object.
- Resolution: Minimum size of the characteristic of the object.
- Depth of Field (PC): The maximum depth of the object that can be kept fully focused.
- Sensor Size: The size of the area of a camera sensor, normally specified in the horizontal dimension.

To obtain the field of view of the camera it is necessary to define the design parameter of the working distance WD $= 1$m and perform calculations based on the following specifications of the Pi v2 camera:

- Vertical field of view $\rightarrow$ vAFOV $= 48.8°$.
- Horizontal field of view $\rightarrow$ hAFOV $= 62.2°$.
- Camera resolution: 3280×2464 px.

The angular field of view (AFOV) is defined in degrees. The calculations of the image field of vision using the vAFOV of the Pi camera (Fig. 6) are as follows:

$$\tan(0.5(\text{vAFOV})) = 0.5\frac{\text{vFOV}}{\text{WD}} \tag{1}$$

Clearing vFOV and replacing the premises we have

$$\text{vFOV} = \frac{100}{0.5}\tan(0.5(48.8°)) = 90.7\text{cm} \tag{2}$$

Now we do the same thing using the hAFOV and we obtain

$$\text{hFOV} = \frac{100}{0.5}\tan(0.5(62.2°)) = 120.6\text{cm} \tag{3}$$

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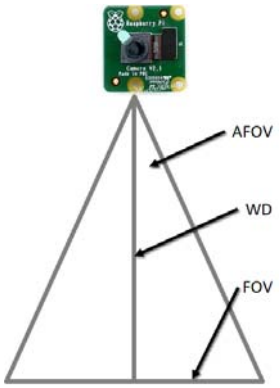


Fig. 6. FOV Camera Pi V2.

Calculating the resolution in pixels per meter (PPM) we have

$$\text{hPPM} = \frac{\text{hPixeles}}{\text{hFOV}} = \frac{3280\text{px}}{1.206\text{m}} \simeq 2719\text{px/m} \quad (4)$$

$$\text{vPPM} = \frac{\text{vPixeles}}{\text{vFOV}} = \frac{2464\text{px}}{0.907\text{m}} \simeq 2716\text{px/m} \quad (5)$$

In order to know the ideal location of the ultrasonic sensor, the field of vision calculations were carried out in the same way as in (1) to (5) considering the sensor specifications:

- Vertical field of view $\rightarrow$ vAFOV$_{SD}$= 15°.
- Horizontal field of view $\rightarrow$ hAFOV$_{SD}$= 15°.

Obtaining the following results:

$$\text{vFOV}_{SD} = 26.33cm$$

$$\text{hFOV}_{SD} = 26.33cm$$

Then, the field of view of both sensors is shown in Fig. 7.

B. Implementation of the vision system for the IoT Sensor SPIN-V

Once that the all components of the SPIN-V are defined, the base base for the raspberry pi, the camera pi and the ultrasonic sensor are designed using SolidWorks and 3D printed. It is

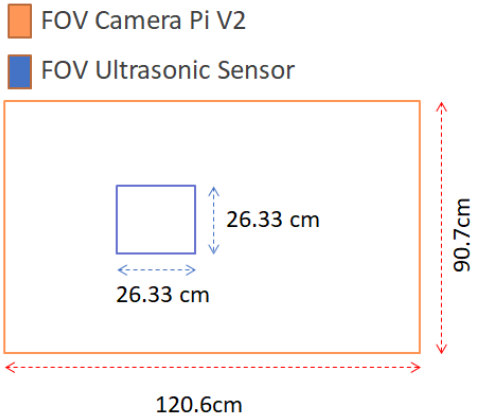


Fig. 7. Field of view of both sensors.

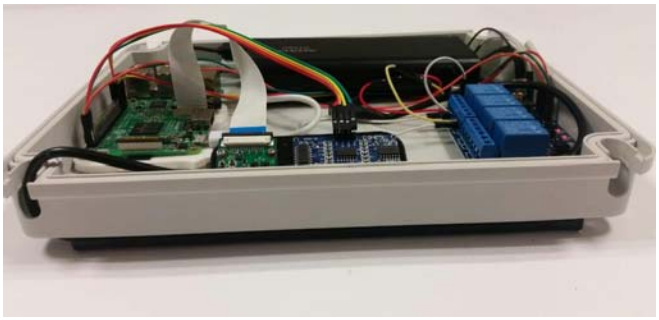


Fig. 8. SPIN-V IoT Sensor prototype.

worthwhile to mentioned that the sensors base mount is designed based on the previously inclination calculations. Once the electronic integration is carried out, the first prototype of the SPIN-V is obtained (see Fig. 8).

In order to evaluate the performance of the SPIN-V, it is mounted in the back end of a parking place of the Universidad del Valle de México campups Guadalajara Sur (Fig. 9).

Once that SPIN-V is mounted at the height established by the design parameters, the image obtained is shown in Fig. 10.

The firmware of the SPIN-V was programmed in Python, using the Python 3 Idle, which is installed in the factory with the Raspian operating system, installed on the Raspberry Pi 3. Several programming languages can be used in the Raspberry Pi. The Python language was chosen due to the following reasons: simplified and Fast, flexible, ordered, portable and libraries included in the codes.

For the detection of the plates, after the photo will be taken, it is converted from image to text in order to upload it to the database. To do that, we worked with OpenCV (Open Source Computer Vision Library), which are libraries that adapt to the Raspberry to make that conversion.



Fig. 9. SPIN-V sensor mounted in the back end of the parking place.

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Fig. 10. Image captured with the SPIN-V.

The complete system has been evaluated with the mobile app, the SPIN-V and the monitoring center obtaining a good performance.

V. Conclusions

The SEI-UVM represents a solution of an intelligent parking system aimed at contributing technologically to the development of smart cities through the use of ICTs. In particular, the IoT sensor SPIN-V has been designed and developed as a compact device and adaptable to different environments since it can be used without the need for expensive infrastructure installation like the solutions of the current guided systems. It should be noted that the electronic solution of the SPIN-V was made with the integration of sensors in the market, so that energy consumption is not optimized and only a functional device was obtained but there are still improvements to be made. Moreover, it is restricted to controlled environment in a private parking lots. Currently, we have three prototype installed in UVM campus Guadalajara Sur monitored from the National Observatory of Smart Environments and we are starting to work with artificial intelligence algorithms applied for parking place suggestion, predictions, among other interesting topics based on this research.

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